

TOPOLOGY GENERAL EXAM FALL 2025

Instructions: This is a four hour exam. Your solutions should be legible and clearly organized, written in complete sentences in good mathematical style on your own paper. All work should be your own-no outside sources are permitted-using methods and results from the first year topology courses. Each problem is worth the same number of points.

Problem 1

Let $X = S^1 \times \mathbb{R}P^2$.

- (a) Compute $\pi_1(X)$.
- (b) Describe the universal cover of X and explicitly describe each element in the group of deck transformations of the universal cover.

Problem 2

Let $x_1, x_2, \dots, x_{10}$ be 10 distinct points on the 2-torus T^2 . Let X be the quotient of T^2 obtained by identifying all 10 points.

- (a) Compute $\pi_1(X)$.
- (b) Compute $H_n(X)$ for all $n \geq 0$.

Problem 3

- (a) Let $f : S^1 \rightarrow \mathbb{R}$ be a continuous map. Show that there exists some $x \in S^1$ such that $f(x) = f(-x)$.
- (b) Show that any continuous map $g : \mathbb{R}P^2 \times \mathbb{R}P^2 \rightarrow T^2$ is nullhomotopic.

Problem 4

Let (C, d) and (C', d') be chain complexes of abelian groups. The mapping cone of a chain map $f : (C, d) \rightarrow (C', d')$, denoted $(\text{cone}(f), d'')$, is defined by

$$\text{cone}(f)_n := C_{n-1} \oplus C'_n, \quad d''(c, c') = (-d(c), d'(c') - f(c)).$$

Prove that $(\text{cone}(f), d'')$ is a chain complex.

Problem 5

Consider the family of smooth maps $f_c : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by

$$f_c(x, y) = (x, y, cx + y^2), \quad c \in \mathbb{R}.$$

Let $S \subset \mathbb{R}^3$ be the surface defined by

$$S = \{(x, y, z) \in \mathbb{R}^3 \mid z = x^2 + y^2\}.$$

- (a) For which values of $c \in \mathbb{R}$ is the map f_c transverse to the surface S ?
- (b) For what values of $c \in \mathbb{R}$ is the intersection of the image of f_c with S a manifold?

Problem 6

Consider the 3-form

$$\theta = e^w(xdy \wedge dz + ydz \wedge dx + zdx \wedge dy)$$

on $\mathbb{R}^4$ where the coordinates are x, y, z, w . Let H be the hemisphere $x^2 + y^2 + z^2 + w^2 = 1, w \geq 0$, with the orientation induced as being part of the boundary of the 4-ball with the standard orientation. Compute

$$\int_S d\theta.$$

Problem 7

Let $p : M \rightarrow N$ be a smooth covering map between closed manifolds of some dimension n . Suppose N is oriented. Show that M is orientable. Pick an orientation on M and compute the degree of p in terms of the number of sheets of the covering.